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IGNITION COIL

Abstract

An ignition coil **2** includes a protector **16** including: a head portion **28** exposed outside an internal combustion engine **4**; and a tubular portion **30** extending downward from a bottom surface of the head portion **28** and inserted in a plug hole **6** of the internal combustion engine **4**. The ignition coil **2** includes an air passage **64** connecting an external environment and the plug hole **6** of the internal combustion engine **4**. The air passage **64** includes a first opening **50** located in a region on an outer surface of a coil assembly **8** located beneath a flange portion **14** or an outer surface of the head portion **28**. The region extends in an up-down direction. When viewed in a reference direction that is perpendicular to a central axis Lf of a mounting hole **20** of the flange portion **14** and extends from the central axis Lf of the mounting hole **20** toward a central axis Lp of the protector **16**, a centerline of the region overlaps with the central axis Lf of the mounting hole **20**. When viewed in the reference direction, a width of the region is equal to a width of the mounting hole **20**.

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Background/Summary

TECHNICAL FIELD

[0001] The present specification discloses an ignition coil for an internal combustion engine.

BACKGROUND ART

[0002] An ignition coil for an internal combustion engine includes: a coil assembly including a coil that generates a high voltage; a protector extending downward from the coil assembly; and a flange portion having a mounting hole. When the ignition coil is mounted on the internal combustion engine, a lower portion of the protector is inserted in a plug hole of the internal combustion engine. A bolt is inserted in the mounting hole of the flange portion, and the ignition coil is fixed to the internal combustion engine via the bolt.

[0003] When the internal combustion engine is in use, the interior of the plug hole becomes hot. The air inside the plug hole expands to increase the pressure inside the plug hole. Japanese Laid-Open Patent Application Publication No. 2010-287485 discloses an ignition device including a communication path for discharging air from a plug hole to the external environment.

CITATION LIST

Patent Literature

[0004] PTL 1: Japanese Laid-Open Patent Application Publication No. 2010-287485

SUMMARY OF INVENTION

Technical Problem

[0005] An internal combustion engine such as an engine mounted in a vehicle is sometimes exposed to rainwater or cleaning water. Entry of this kind of water into a plug hole of the internal combustion engine through an air passage connecting the external environment and the plug hole could disturb proper combustion. There is a demand for an ignition coil that can reduce water entry into a plug hole.

[0006] The present inventors aim to provide an ignition coil including an air passage that is resistant to water entry.

Solution to Problem

[0007] An ignition coil for an internal combustion engine according to one embodiment includes: a coil assembly including a primary coil and a secondary coil; an output portion extending downward from the coil assembly; a protector covering an outer circumference of the output portion and extending downward from the coil assembly; and a flange portion projecting backward from the coil assembly and including a mounting hole extending from top to bottom of the flange portion. The protector includes: a head portion that is exposed outside the internal combustion engine when the ignition coil is mounted on the internal combustion engine; and a tubular portion that extends downward from a bottom surface of the head portion and that is inserted in a plug hole of the internal combustion engine when the ignition coil is mounted on the internal combustion engine. The ignition coil includes an air passage connecting an external environment and the plug hole of the internal combustion engine. The air passage includes a first opening located in a region on an outer surface of the coil assembly located beneath the flange portion or an outer surface of the head portion. The region extends in an up-down direction. When viewed in a reference direction that is perpendicular to a central axis of the mounting hole and extends from the central axis of the mounting hole toward a central axis of the protector, a centerline of the region overlaps with the central axis of the mounting hole. When viewed in the reference direction, a width of the region is

equal to a width of the mounting hole.

Advantageous Effects of Invention

[0008] The ignition coil of this embodiment includes the air passage connecting the external environment and the plug hole of the internal combustion engine. The first opening of the air passage is located in the region located on an outer surface of the coil assembly beneath the flange portion or an outer surface of the head portion. The region extends in the up-down direction. When viewed in a reference direction that is perpendicular to a central axis of the mounting hole and extends from the central axis of the mounting hole toward a central axis of the protector, a centerline of the region overlaps with the central axis of the mounting hole. When viewed in the reference direction, a width of the region is equal to a width of the mounting hole. When the ignition coil is mounted on the internal combustion engine, a bolt is inserted in the mounting hole. The bolt or a receiving portion that receives the bolt is located in a direction in which the region faces. External water is blocked by the bolt or the receiving portion and cannot readily reach the first opening. The ignition coil is resistant to water entry into the air passage.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1A is a perspective view showing an ignition coil according to one embodiment, and FIG. 1B is a cross-sectional view of the ignition coil of FIG. 1A.

[0010] FIG. 2 is an enlarged cross-sectional view of a part of FIG. 1B.

[0011] FIG. 3 is an enlarged back view of a part of the ignition coil of FIG. 1.

[0012] FIG. 4A is a plan view of a protector of the ignition coil of FIG. 1, and FIG. 4B is a bottom view of the protector.

[0013] FIG. 5 is an enlarged cross-sectional view of a part of FIG. 2.

[0014] FIG. 6A is a back view showing a part of an ignition coil according to another embodiment, and FIG. 6B is a perspective view showing a part of the ignition coil of FIG. 6A.

[0015] FIG. 7 is an enlarged plan view of a part of a protector of the ignition coil of FIG. 6A.

[0016] FIG. 8A is a perspective view showing a part of an ignition coil according to yet another embodiment, FIG. 8B is a perspective view showing a part of a coil assembly of the ignition coil of FIG. 8A, and FIG. 8C is a perspective view showing a part of a protector of the ignition coil of FIG. 8A.

[0017] FIG. 9 is a perspective view showing a part of an ignition coil according to yet another embodiment.

[0018] FIG. 10A is a perspective view showing a part of an ignition coil according to yet another embodiment, FIG. 10B is a perspective view showing a part of a coil assembly of the ignition coil of FIG. 10A, and FIG. 10C is a perspective view showing a part of a protector of the ignition coil of FIG. 10A.

[0019] FIG. 11 is a perspective view showing a part of an ignition coil according to yet another embodiment.

DESCRIPTION OF EMBODIMENTS

[0020] The following will describe in detail preferred embodiments with appropriate reference to the drawings.

[0021] FIG. 1A is a perspective view showing an ignition coil 2 according to one embodiment. In FIG. 1, the arrow X represents the forward direction with respect to the ignition coil 2. The opposite direction is the backward direction. The arrow Y represents the rightward direction with respect to the ignition coil 2. The opposite direction is the leftward direction. The arrow Z represents the upward direction with respect to the ignition coil 2. The opposite direction is the downward direction. The direction-indicating terms such as “forward” and “upward” are used

herein to express the relative positional relationship among the below-described components of the ignition coil **2**. For example, the direction shown as the “backward” direction in FIG. **1** may be regarded as a “forward” or “rightward” direction as long as the relative positional relationship can be expressed.

[0022] FIG. **1B** is a cross-sectional view of the ignition coil **2** of FIG. **1A** taken along a plane perpendicular to the left-right direction. FIG. **1B** further shows an internal combustion engine **4** with the ignition coil **2** mounted thereon. The internal combustion engine **4** includes a plug hole **6**, and a part of the ignition coil **2** is inserted in the plug hole **6**. FIG. **2** is an enlarged view of a part of FIG. **1B**. The ignition coil **2** includes a coil assembly **8**, a connector portion **10**, an output portion **12**, a flange portion **14**, and a protector **16**. In FIG. **1B** and FIG. **2**, the internal structures of the coil assembly **8**, the connector portion **10**, and the output portion **12** are omitted.

[0023] The coil assembly **8** includes: a primary coil, a secondary coil, and an iron core which are not shown in the drawings; and a case enclosing the coils and the iron core. The primary coil is formed by winding a wire around the iron core and the secondary coil is formed by winding a wire around the outside of the primary coil. The number of wire turns in the secondary coil is much greater than the number of wire turns in the primary coil. Thus, a change in the current flowing through the primary coil can generate a high voltage in the secondary coil.

[0024] The connector portion **10** is located ahead of the coil assembly **8**. The connector portion **10** includes an external terminal **18** to which an external signal is input and an unshown ignitor. The ignitor is a switch that permits or prevents a current flow through the primary coil in response to the signal input to the external terminal **18**. The operation of the ignition coil **2** is controlled by the signal input to the external terminal **18**.

[0025] As shown in FIG. **2**, the output portion **12** is located below the coil assembly **8**. The output portion **12** is shaped as a tube extending downward from the coil assembly **8**. The high voltage generated in the secondary coil is supplied through the output portion **12** to an ignition plug mounted on the internal combustion engine **4**.

[0026] The flange portion **14** projects backward from the coil assembly **8**. The flange portion **14** extends in a direction perpendicular to a direction in which the output portion **12** extends. The flange portion **14** is integral with the case of the coil assembly **8**. The flange portion **14** includes a mounting hole **20** extending from top to bottom of the flange portion **14**. As shown in FIG. **1B** and FIG. **2**, a receiving portion **22** of the internal combustion engine **4** is located beneath the mounting hole **20**. The receiving portion **22** includes a mounting hole **24**. When the ignition coil **2** is mounted on the internal combustion engine **4**, a bolt **25** is inserted in the mounting hole **20** of the flange portion **14** and the mounting hole **24** of the receiving portion **22**. FIG. **1B** shows the bolt **25**. The ignition coil **2** is fixed to the internal combustion engine **4** by the bolt **25**. The bolt **25** is omitted in FIG. **2**.

[0027] In this embodiment, the connector portion **10** is located ahead of the coil assembly **8** and the flange portion **14** is located behind the coil assembly **8**. The flange portion **14** is across the coil assembly **8** from the connector portion **10**. The connector portion **10** and the flange portion **14** may be located, respectively, along side surfaces of the coil assembly **8** that are perpendicular to each other. For example, the connector portion **10** may be located on the right of the coil assembly **8** while the flange portion **14** is located behind the coil assembly **8**. The connector portion **10** may be located in another direction relative to the flange portion **14**. The direction in which the flange portion **14** is located relative to the coil assembly **8** is defined herein as the “backward” direction with respect to the ignition coil **2**, regardless of factors such as the positional relationship between the connector portion **10** and the flange portion **14** and the shape of the coil assembly **8**. The direction in which the output portion extends relative to the coil assembly **8** is defined herein as the “downward” direction with respect to the ignition coil **2**.

[0028] The protector **16** is located below the coil assembly **8**. The protector **16** includes a hole extending through the interior of the protector **16**. The protector **16** is fitted around the output

portion **12**. The protector **16** covers the outer circumference of the output portion **12**. The protector **16** is made of a rubber composition. A preferred example of the material of the protector **16** is silicone rubber. The protector **16** includes a head portion **28** and a tubular portion **30**.
[0029] The head portion **28** is ring-shaped. The head portion **28** includes a hole **32** extending through the central region of the head portion **28**. The output portion **12** is inserted in the hole **32** of the head portion **28**. The output portion **12** extends through the hole **32** of the head portion **28** and extends further downward from the head portion **28**. The inner circumferential surface of the hole **32** of the head portion **28** is in contact with the outer circumferential surface of the output portion **12**, without any gap between the inner and outer circumferential surfaces. The top surface **34** of the head portion **28** is in contact with the bottom surface **36** of the coil assembly **8**.

[0030] As shown in FIG. **1B** and FIG. **2**, when the ignition coil **2** is mounted on the internal combustion engine **4**, the head portion **28** is exposed outside the internal combustion engine **4**. The head portion **28** covers the opening of the plug hole **6** of the internal combustion engine **4**. The outer diameter of the head portion **28** is greater than the inner diameter of the plug hole **6**. The inner diameter of the head portion **28** is smaller than the inner diameter of the plug hole **6**. As shown in FIG. **2**, the bottom surface **38** of the head portion **28** includes a fitting groove **40**. The fitting groove **40** is ring-shaped (see FIG. **4B**). The top surface of the internal combustion engine **4** includes a ring-shaped protrusion **42**. As shown in FIG. **2**, the protrusion **42** of the internal combustion engine **4** is fitted in the fitting groove **40** of the head portion **28**. The head portion **28** serves as a lid closing the opening of the plug hole **6**.

[0031] The tubular portion **30** extends downward from the head portion **28**. In this embodiment, the tubular portion **30** is integral with the head portion **28**. The tubular portion **30** and the head portion **28** may be formed separately from each other. The tubular portion **30** includes a hole **44** continuous with the hole **32** of the head portion **28**. The outer diameter of the tubular portion **30** is smaller than the outer diameter of the head portion **28**. The outer diameter of the tubular portion **30** is smaller than the inner diameter of the plug hole **6**. As shown in FIG. **1B** and FIG. **2**, when the ignition coil **2** is mounted on the internal combustion engine **4**, the tubular portion **30** is inserted in the plug hole **6**. There is a gap between the outer circumferential surface **46** of the tubular portion **30** and the inner circumferential surface **48** of the internal combustion engine **4** that defines the plug hole **6**.

[0032] In FIG. **2**, the dashed-dotted line Lf represents the central axis of the mounting hole **20**. The dashed-dotted line Lp represents the central axis of the protector **16**. The central axis Lp of the protector **16** extends in the up-down direction and passes through the center of the hole **32** of the head portion **28** at the level of the bottom surface **38** of the head portion **28**. In FIG. **2**, the arrow D represents a direction perpendicular to the central axis Lf and extending from the central axis Lf toward the central axis Lp. This direction is referred to herein as a reference direction D. In this embodiment, the reference direction D is the same as the direction X extending from back to front of the ignition coil **2**.

[0033] FIG. **3** shows a part of the ignition coil **2** as viewed in the reference direction D. In this embodiment, FIG. **3** is a back view of the ignition coil **2**. In this figure, the central axis Lf and the central axis Lp coincide with each other. In FIG. **3**, the reference sign R represents a region located beneath the flange portion **14** and lying on the outer surfaces of the coil assembly **8** and the head portion **28**. The region R extends from the lower end of the flange portion **14** to the lower end of the head portion **28** in the up-down direction. The region R, as viewed in the reference direction D, has a centerline Lr overlapping with the central axis Lf of the mounting hole **20** and has a width equal to the width of the mounting hole **20**. This region R is referred to herein as a “location region R”.

[0034] The ignition coil **2** includes an opening **50** in the location region R. The expression “includes an opening in the location region R” means that a part or all of the opening **50** is located within the location region R. As described later, the opening **50** is a first opening **50** of an air passage of the ignition coil **2**. In the embodiment of FIG. **3**, the first opening **50** is located at an end

of the head portion **28** that is adjacent to the coil assembly **8**. In the embodiment of FIG. 3, the first opening **50** is located on the centerline **Lr**.

[0035] FIG. 4A is a plan view of the protector **16**. This figure shows the top surface **34** of the head portion **28**. FIG. 4B is a bottom view of the protector **16**. This figure shows the bottom surface **38** of the head portion **28** and the tubular portion **30**. FIG. 5 is an enlarged cross-sectional view of the head portion **28** and its vicinity.

[0036] As shown in FIG. 4A and FIG. 5, the top surface **34** of the head portion **28** includes a lead-out groove **52** extending from the first opening **50**, a first ring groove **54**, and a second ring groove **56**. As shown in FIG. 4A, the top surface **34** of the head portion **28** further includes a connection groove **58**. The lead-out groove **52** connects the first opening **50** and the first ring groove **54**. The connection groove **58** connects the first ring groove **54** and the second ring groove **56**. As shown in FIG. 4A, the bottom of the second ring groove **56** includes a hole opening **59**. As shown in FIG. 4B, the bottom surface **38** of the head portion **28** includes a hole opening **60**. As shown in FIG. 5, there is an air hole **62** extending from the bottom of the second ring groove **56** to the bottom surface **38** of the head portion **28**. The opening **60** in the bottom surface **38** of the head portion **28** is a second opening **60** of the air passage. In other words, the ignition coil **2** includes an air passage **64** extending from the first opening **50** to the second opening **60** through the lead-out groove **52**, the first ring groove **54**, the connection groove **58**, the second ring groove **56**, and the air hole **62**. In FIG. 4A and FIG. 5, the air passage **64** is shown by a dashed line.

[0037] As shown FIG. 4B, the fitting groove **40** in the bottom surface **38** of the head portion **28** is ring-shaped. The second opening **60** is located inside the fitting groove **40** and outside the tubular portion **30**. As shown in FIG. 5, the second opening **60** is located between the tubular portion **30** and the inner circumferential surface of the internal combustion engine **4** which defines the plug hole **6**. The second opening **60** is connected to the plug hole **6**. That is, the air passage **64** of the ignition coil **2** connects the external environment and the plug hole **6**.

[0038] The following describes the effects and benefits of the present embodiment.

[0039] The ignition coil **2** of this embodiment includes the air passage **64** connecting the external environment and the plug hole **6** of the internal combustion engine **4**. The first opening **50** of the air passage **64** is located in the location region **R**. When the ignition coil **2** is mounted on the internal combustion engine **4**, the bolt **25** is inserted in the mounting hole **20** of the flange portion **14**. The receiving portion **22** that receives the bolt **25** is located beneath the mounting hole **20**. The outer diameter of the receiving portion **22** is greater than the inner diameter of the mounting hole **20**. The receiving portion **22** is located in the direction in which the location region **R** faces, and the flange portion **14** is located above the location region **R**. External water is blocked by the receiving portion **22** and the flange portion **14** and cannot readily reach the first opening **50**. The ignition coil **2** is resistant to water entry into the air passage **64**.

[0040] In some cases, there is not the receiving portion **22**, and the bolt **25** inserted in the mounting hole **20** extends to the top surface of the internal combustion engine. In such a case, the bolt **25** is located in the direction in which the location region **R** faces. External water is blocked by the bolt **25**. Since the first opening **50** is located in the location region **R**, external water is blocked by the receiving portion **22** or the bolt **25**.

[0041] In FIG. 3, the reference sign **Wr** represents the width of the location region **R**. In view of effectively preventing water entry into the air passage **64**, the distance between the center of the first opening **50** and the centerline **Lr** is preferably 30% or less of the width **Wr**, more preferably 20% or less of the width **Wr**, and even more preferably 10% or less of the width **Wr**. Most preferably, the center of the first opening **50** is located on the centerline **Lr**.

[0042] The first opening **50** in the location region **R** is preferably located at an end of the head portion **28** that is adjacent to the coil assembly **8**. In this case, the air passage **64** can be formed by grooves in the top surface **34** of the head portion **28** and the air hole **62** extending from the bottom of one of the grooves to the bottom surface **38** of the head portion **28**. In this embodiment, the air

passage **64** is formed by the lead-out groove **52**, the first ring groove **54**, the connection groove **58**, the second ring groove **56**, and the air hole **62**. The formation of the air passage **64** does not require any hole extending through the interior of the head portion **28** or the coil assembly **8** in a horizontal direction. This makes the production of the ignition coil **2** easy.

[0043] The second opening **60** is preferably located in the bottom surface **38** of the head portion **28**. Locating the second opening **60** in the bottom surface **38** of the head portion **28** can reduce the length of the air hole **62**. This makes the production of the ignition coil **2** easy.

[0044] The location of the second opening **60** is not limited to the bottom surface **38** of the head portion **28**. For example, the second opening **60** may be located in the outer circumferential surface **46** of the tubular portion **30**. In this case, for example, the second ring groove and the air hole's opening in the second ring groove are located above the tubular portion **30**. The air hole extends from the bottom of the second ring groove, passes through the head portion **28** and a part of the tubular portion **30**, and reaches the outer circumferential surface **46** of the tubular portion **30**. The second opening **60** may be at any location where the second opening **60** is connected to the plug hole **6**.

[0045] FIG. **6A** is a back view showing a part of an ignition coil **70** according to another embodiment, and FIG. **6B** is a perspective view showing a part of the ignition coil **70**. As shown in FIGS. **6A** and **6B**, the ignition coil **70** includes a coil assembly **72**, a flange portion **74**, and a protector **76**. The ignition coil **70** includes a guard **80** located around a first opening **78**. The ignition coil **70** is the same as the ignition coil **2** of FIG. **1**, except for including the guard **80**.

[0046] FIG. **7** is a plan view showing a part of a head portion **82** of the protector **76** of FIG. **6**. In this embodiment, the guard **80** projects from a side surface **83** of the head portion **82**. The guard **80** is located on both sides of the first opening **78** (in the circumferential direction of the head portion **82**). In this embodiment, the guard **80** includes a first portion **80a** located on one side of the first opening **78** and a second portion **80b** located on the other side of the first opening **78**. The first portion **80a** and the second portion **80b** are columnar.

[0047] FIG. **7** further shows a cross-section **88** of a bolt **86** inserted in a mounting hole **84** of the flange portion **74**. The bolt **86** has an outer diameter equal to the inner diameter of the mounting hole **84** and extends downward beyond the first opening **78**. In reality, the receiving portion **22** of the internal combustion engine **4**, which has a greater outer diameter than the bolt **86**, is located in the direction in which the mounting hole **84** faces. That is, the cross-section **88** is the smallest of the cross-sections of components that may be located in the direction in which the first opening **78** faces. In FIG. **7**, the dashed line **V1** is a line tangent to the outer circumferential surface of the bolt **86** and passing through a corner of that surface of the first portion **80a** which is adjacent to the first opening **78**, and the dashed line **V2** is a line tangent to the outer circumferential surface of the bolt **86** and passing through a corner of that surface of the second portion **80b** which is adjacent to the first opening **78**. Neither of the tangent lines **V1** and **V2** reach the first opening **78**. That is, the first opening **78** is hidden behind the bolt **86** or the guard **80** when the ignition coil **70** is viewed at the level of the first opening **78** in any direction perpendicular to the up-down direction. The first opening **78** in such a state is described herein as being "in a hidden state". In other words, the height of the guard **80** (the height of each of the first and second portions **80a** and **80b** in this embodiment) is such that the first opening **78** is in a hidden state.

[0048] The ignition coil **70** of this embodiment includes the guard **80** located on both sides of the first opening **78** and projecting from the side surface **83** of the head portion **82**. The guard **80** effectively prevents external water from reaching the first opening **78** sideways. The ignition coil **70** is resistant to water entry into the air passage.

[0049] As previously stated, the height of the guard **80** is preferably such that the first opening **78** is in a hidden state. In this case, the guard **80** more effectively prevents external water from reaching the first opening **78**. The ignition coil **70** is resistant to water entry into the air passage.

[0050] The height of the guard **80** need not be such that the first opening **78** is in a hidden state.

The height of the guard **80** may be smaller or greater than the minimum height which is such that the first opening **78** is in a hidden state. However, the height of the guard **80** needs to be such that the guard **80** does not contact the receiving portion **22**.

[0051] The guard **80** is not limited to the shape as shown in FIG. 7. The first and second portions **80a** and **80b** may become thinner as they extend outward. Each of the first and second portions **80a** and **80b** may be shaped as an arc projecting from the side surface **83** of the head portion **82**. The guard **80** may include a third portion located below the first opening **78**, projecting outward, and connecting the first portion **80a** and the second portion **80b**. The guard **80** may be shaped as a ring projecting from the side surface **83** of the head portion **82** and a side surface of the coil assembly **72** and surrounding the first opening **78**. The guard **80** may be in any other shape. It is sufficient that the guard **80** be located on both sides of the first opening **78** and project from the side surface **83** of the head portion **82**.

[0052] FIG. **8A** is a perspective view showing a part of an ignition coil **90** according to yet another embodiment, FIG. **8B** is a perspective view of the ignition coil **90** with a protector **94** removed from the ignition coil **90**, and FIG. **8C** is a perspective view of the removed protector **94**. FIG. **8B** partially shows a coil assembly **92** and an output portion **96** of the ignition coil **90**.

[0053] As shown in FIG. **8B**, the ignition coil **90** includes a recess **100** recessed upward and located at an end of a bottom surface **98** of the coil assembly **92**, the end of the bottom surface **98** being adjacent to a side surface **99** of the coil assembly **92**. As shown in FIG. **8C**, there is a protrusion **106** extending upward and located at an end of a top surface **104** of a head portion **102** of the protector **94**, the end of the top surface **104** being adjacent to a side surface **105** of the head portion **102**. As shown in FIG. **8A**, when the coil assembly **92** and the protector **94** are assembled together, the protrusion **106** is fitted in the recess **100**. The ignition coil **90** is the same as the ignition coil **2** of FIG. **1**, except for including the recess **100** and the protrusion **106** fitted in the recess **100**.

[0054] In the ignition coil **90**, since the protrusion **106** of the head portion **102** is fitted in the recess **100** of the coil assembly **92**, the protector **94** is prevented from rotating relative to the coil assembly **92** during assembly of the coil assembly **92** and the protector **94**. This allows for accurate positioning of a first opening **108** relative to the coil assembly **92**. Since the case of the coil assembly **92** and a flange portion **110** are integral with each other, the first opening **108** can be accurately positioned relative to a central axis L_f of a mounting hole **112** of the flange portion **110**. In the ignition coil **90**, the first opening **108** can be accurately positioned on the centerline L_r of the location region **R**. The ignition coil **90** is resistant to water entry into the air passage.

[0055] Although not shown, an ignition coil according to yet another embodiment includes: a protrusion extending downward and located at an end of a bottom surface of a coil assembly, the end of the bottom surface being adjacent to a side surface of the coil assembly; and a recess recessed downward and located at an end of a top surface of a head portion of a protector, the end of the top surface being adjacent to a side surface of the head portion. When the coil assembly and the protector are assembled together, the protrusion is fitted in the recess. The ignition coil is the same as the ignition coil **2** of FIG. **1**, except for including the recess and the protrusion fitted in the recess. In the ignition coil, the protector is prevented from rotating relative to the coil assembly during assembly of the coil assembly and the protector.

[0056] FIG. **9** is a perspective view showing a part of an ignition coil **120** according to yet another embodiment. The ignition coil **120** includes a recess **124** located in a bottom surface of a coil assembly **122** and a protrusion **130** located at an upper edge of a head portion **128** of a protector **126**, the protrusion **130** being fitted in the recess **124**. In this embodiment, the protrusion **130** of the head portion **128** and the recess **124** of the coil assembly **122** are located in the location region **R**. A first opening **132** is located in a side surface **131** of the protrusion **130** of the head portion **128**. The ignition coil **120** is the same as the ignition coil **90** of FIG. **8**, except that the protrusion **130** of the head portion **128** and the recess **124** of the coil assembly **122** are located in the location region **R** and that the first opening **132** is located in the side surface **131** of the protrusion **130**.

[0057] In the ignition coil **120**, the first opening **132** is located in the side surface **131** of that protrusion **130** of the head portion **128** which is fitted in the recess **124** of the coil assembly **122**. This allows for accurate positioning of the first opening **132** relative to the coil assembly **122**. The first opening **132** can be accurately positioned relative to a central axis Lf of a mounting hole **136** of a flange portion **134**. The ignition coil **120** is resistant to water entry into the air passage.

[0058] FIG. **10A** is a perspective view showing a part of an ignition coil **140** according to yet another embodiment, FIG. **10B** is a perspective view of the ignition coil **140** with a protector **144** removed from the ignition coil **140**, and FIG. **10C** is a perspective view of the removed protector **144**. FIG. **10B** partially shows a coil assembly **142** and an output portion **146** of the ignition coil **140**.

[0059] In this embodiment, as shown in FIG. **10A**, a first opening **148** is located in a side surface **143** of the coil assembly **142**. The first opening **148** is located at an end of the side surface **143** of the coil assembly **142**, the end of the side surface **143** being adjacent to a head portion **150** of the protector **144**. As shown in FIG. **10B**, a bottom surface **152** of the coil assembly **142** includes a lead-out groove **154** extending from the first opening **148**. Although not shown, a top surface **156** of the head portion **150** includes a first ring groove, a connection groove, and a second ring groove, as with the head portion **28** of FIG. **4A**. Furthermore, there is an air hole extending from the bottom of the second ring groove to a bottom surface **158** of the head portion **150**. The lead-out groove **154** in the bottom surface **152** of the coil assembly **142** is connected to the first ring groove. In this embodiment, an air passage is formed by the lead-out groove **154**, the first ring groove, the connection groove, the second ring groove, and the air hole.

[0060] In this embodiment, the first opening **148** is located in the location region R and at that end of the side surface **143** of the coil assembly **142** which is adjacent to the head portion **150**. Thus, the air passage can be formed by the groove located in the bottom surface **152** of the coil assembly **142**, the grooves located in the top surface **156** of the head portion **150**, and the air hole extending from the bottom of one of the grooves to the bottom surface **158** of the head portion **150**. The formation of the air passage does not require any hole extending through the interior of the head portion **150** or the coil assembly **142** in a horizontal direction. This makes the production of the ignition coil **140** easy.

[0061] In this embodiment, the first opening **148** is located in the side surface **143** of the coil assembly **142**. The position of the first opening **148** relative to a central axis Lf of a mounting hole **162** of a flange portion **160** is not affected by rotation of the protector **144** relative to the coil assembly **142**. The first opening **148** can be accurately positioned relative to the central axis Lf of the mounting hole **162** of the flange portion **160**. In the ignition coil **140**, the first opening **148** can be accurately positioned on the centerline Lr of the location region R. The ignition coil **140** is resistant to water entry into the air passage.

[0062] Although not shown, an ignition coil according to yet another embodiment includes: a protrusion extending downward and located at an end of a bottom surface of a coil assembly, the end of the bottom surface being adjacent to a side surface of the coil assembly; a recess recessed downward and located at an end of a top surface of a head portion of a protector, the end of the top surface being adjacent to a side surface of the head portion; and a first opening located in a side surface of the protrusion of the coil assembly. When the coil assembly and the protector are assembled together, the protrusion is fitted in the recess. The ignition coil is the same as the ignition coil **140** of FIG. **10**, except that that the ignition coil includes the recess and the protrusion fitted in the recess and that the first opening is located in the side surface of the protrusion.

[0063] FIG. **11** is a perspective view showing a part of an ignition coil **170** according to yet another embodiment. In the ignition coil **170**, the first opening is located in the location region R and at an end of a side surface **178** of a coil assembly **172**, the end of the side surface being adjacent to a head portion **178** of a protector **176**. A guard **174** is located around the first opening. The ignition coil **170** is the same as the ignition coil **140** of FIG. **10**, except that the coil assembly **172** includes

the guard **174**.

[0064] In this embodiment, the guard **174** projects from the side surface **178** of the coil assembly **172**. The guard **174** is located on both sides of the first opening. In this embodiment, the guard **174** includes a first portion **174a** located on one side of the first opening and a second portion **174b** located on the other side of the first opening.

[0065] The guard **174** is not limited to the shape as shown in FIG. **11**. Each of the first and second portions **174a** and **174b** may be shaped as an arc projecting from the side surface **178** of the coil assembly **172**. The guard **174** may include a third portion located above the first opening, projecting outward, and connecting the first portion **174a** and the second portion **174b**. The guard **174** may be shaped as a ring projecting from the side surface **178** of the coil assembly **172** and a side surface **180** of the head portion **178** and surrounding the first opening. The guard **174** may be in any other shape. It is sufficient that the guard **174** be located on both sides of the first opening and project from the side surface **178** of the coil assembly **172**.

[0066] In the embodiments described above, the first opening is located either in the side surface of the head portion or in the side surface of the coil assembly. The first opening may extend from the side surface of the head portion to the side surface of the coil assembly. In this case, the top surface of the head portion includes a lead-out groove extending from the first opening, and the bottom surface of the coil assembly includes a lead-out groove extending from the first opening.

[0067] According to any of the present embodiments, as described above, water entry into the air passage of the ignition coil can be reduced. This demonstrates the superiority of the present embodiments.

Disclosed Items

[0068] The following items disclose preferred embodiments.

[Item 1]

[0069] An ignition coil for an internal combustion engine, the ignition coil including: [0070] a coil assembly including a primary coil and a secondary coil; [0071] an output portion extending downward from the coil assembly; [0072] a protector covering an outer circumference of the output portion and extending downward from the coil assembly; and [0073] a flange portion projecting backward from the coil assembly and including a mounting hole extending from top to bottom of the flange portion, wherein [0074] the protector includes [0075] a head portion that is exposed outside the internal combustion engine when the ignition coil is mounted on the internal combustion engine, and [0076] a tubular portion that extends downward from a bottom surface of the head portion and that is inserted in a plug hole of the internal combustion engine when the ignition coil is mounted on the internal combustion engine, [0077] the ignition coil includes an air passage connecting an external environment and the plug hole of the internal combustion engine, [0078] the air passage includes a first opening located in a region on an outer surface of the coil assembly located beneath the flange portion or an outer surface of the head portion, [0079] the region extends in an up-down direction, [0080] when viewed in a reference direction that is perpendicular to a central axis of the mounting hole and extends from the central axis of the mounting hole toward a central axis of the protector, a centerline of the region overlaps with the central axis of the mounting hole, and [0081] when viewed in the reference direction, a width of the region is equal to a width of the mounting hole.

[Item 2]

[0082] The ignition coil according to item 1, wherein the air passage includes a second opening located in the bottom surface of the head portion.

[Item 3]

[0083] The ignition coil according to item 1 or 2, including a guard located on both sides of the first opening and projecting from a side surface of the head portion or a side surface of the coil assembly.

[Item 4]

[0084] The ignition coil according to item 3, wherein [0085] the guard has a height such that, when a bolt having an outer diameter equal to an inner diameter of the mounting hole and extending downward beyond the first opening is inserted in the mounting hole and the ignition coil is viewed at a level of the first opening in any direction perpendicular to the up-down direction, the first opening is hidden behind the rod member or the guard.

[Item 5]

[0086] The ignition coil according to any one of items 1 to 4, wherein [0087] the head portion includes a protrusion located at an end of a top surface of the head portion and extending upward from the top surface, the end of the top surface being adjacent to a side surface of the head portion, and [0088] the coil assembly includes a recess which is located in a bottom surface of the coil assembly and in which the protrusion is fitted.

[Item 6]

[0089] The ignition coil according to item 5, wherein the first opening is located in a side surface of the protrusion.

[Item 7]

[0090] The ignition coil according to any one of items 1 to 6, wherein [0091] the first opening is located at an end of a side surface of the head portion, the end of the side surface being adjacent to the coil assembly, and [0092] the air passage includes [0093] a groove located in a top surface of the head portion and connected to the first opening, and [0094] a hole extending from a bottom of the groove to the second opening.

[Item 8]

[0095] The ignition coil according to any one of items 1 to 7, wherein [0096] the first opening is located at an end of a side surface of the coil assembly, the end of the side surface being adjacent to the head portion.

[Item 9]

[0097] The ignition coil according to item 8, wherein [0098] the air passage includes a second opening located in the bottom surface of the head portion, and [0099] the air passage further includes [0100] a first groove located in a bottom surface of the coil assembly and connected to the first opening, [0101] a second groove located in a top surface of the head portion and connected to the first groove, and [0102] a hole extending from a bottom of the second groove to the second opening.

[Item 10]

[0103] An ignition coil-internal combustion engine assembly including: [0104] the ignition coil according to any one of items 1 to 9; and [0105] an internal combustion engine including a plug hole in which the tubular portion of the protector of the ignition coil is inserted.

INDUSTRIAL APPLICABILITY

[0106] The ignition coil as described above is used in various internal combustion engines.

REFERENCE SIGNS LIST

[0107] **2, 70, 90, 120, 140, 170** . . . ignition coil [0108] **4** . . . internal combustion engine [0109] **6** . . . plug hole [0110] **8, 72, 92, 122, 142, 172** . . . coil assembly [0111] **10** . . . connector portion [0112] **12, 96, 146** . . . output portion [0113] **14, 74, 110, 134, 160** . . . flange portion [0114] **16, 76, 94, 126, 144, 176** . . . protector [0115] **20, 84, 112, 136, 162** . . . mounting hole of flange portion [0116] **22** . . . receiving portion [0117] **24** . . . mounting hole of receiving portion [0118] **28, 82, 102, 128, 150, 178** . . . head portion [0119] **30** . . . tubular portion [0120] **32** . . . hole of head portion [0121] **34, 104, 156** . . . top surface of head portion [0122] **36, 98** . . . bottom surface of coil assembly [0123] **38, 158** . . . bottom surface of head portion [0124] **40** . . . fitting groove [0125] **42** . . . protrusion of internal combustion engine [0126] **44** . . . hole of tubular portion [0127] **46** . . . outer circumferential surface of tubular portion [0128] **48** . . . inner circumferential surface [0129] **50, 78, 108, 132, 148** . . . first opening [0130] **52, 154** . . . lead-out groove [0131] **54** . . . first ring groove [0132] **56** . . . second ring groove [0133] **58** . . . connection groove [0134] **60** . . . second

opening [0135] **62** . . . air hole [0136] **64** . . . air passage [0137] **80, 174** . . . guard [0138] **83, 105, 180** . . . side surface of head portion [0139] **25, 86** . . . bolt [0140] **88** . . . cross-section [0141] **99, 143, 178** . . . side surface of coil assembly [0142] **100, 124** . . . recess [0143] **106, 130** . . . protrusion of head portion [0144] **131** . . . side surface of protrusion of head portion

Claims

1. An ignition coil for an internal combustion engine, the ignition coil comprising: a coil assembly including a primary coil and a secondary coil; an output portion extending downward from the coil assembly; a protector covering an outer circumference of the output portion and extending downward from the coil assembly; and a flange portion projecting backward from the coil assembly and including a mounting hole extending from top to bottom of the flange portion, wherein the protector includes a head portion that is exposed outside the internal combustion engine when the ignition coil is mounted on the internal combustion engine, and a tubular portion that extends downward from a bottom surface of the head portion and that is inserted in a plug hole of the internal combustion engine when the ignition coil is mounted on the internal combustion engine, the ignition coil includes an air passage having a first opening and a second opening and connecting an external environment and the plug hole of the internal combustion engine, the first opening is located in a region on an outer surface of the coil assembly located beneath the flange portion or an outer surface of the head portion, the region extends in an up-down direction, when viewed in a reference direction that is perpendicular to a central axis of the mounting hole and extends from the central axis of the mounting hole toward a central axis of the protector, a centerline of the region overlaps with the central axis of the mounting hole, and when viewed in the reference direction, a width of the region is equal to a width of the mounting hole.
2. The ignition coil according to claim 1, wherein the second opening is located in the bottom surface of the head portion.
3. The ignition coil according to claim 1, comprising a guard located on both sides of the first opening and projecting from a side surface of the head portion or a side surface of the coil assembly.
4. The ignition coil according to claim 3, wherein the guard has a height such that, when a bolt having an outer diameter equal to an inner diameter of the mounting hole and extending downward beyond the first opening is inserted in the mounting hole and the ignition coil is viewed at a level of the first opening in any direction perpendicular to the up-down direction, the first opening is hidden behind the bolt or the guard.
5. The ignition coil according to claim 1, wherein the head portion includes a protrusion located at an end of a top surface of the head portion and extending upward from the top surface, the end of the top surface being adjacent to a side surface of the head portion, and the coil assembly includes a recess which is located in a bottom surface of the coil assembly and in which the protrusion is fitted.
6. The ignition coil according to claim 5, wherein the first opening is located in a side surface of the protrusion.
7. The ignition coil according to claim 1, wherein the first opening is located at an end of a side surface of the head portion, the end of the side surface being adjacent to the coil assembly, and the air passage includes a groove located in a top surface of the head portion and connected to the first opening, and a hole extending from a bottom of the groove to the second opening.
8. The ignition coil according to claim 1, wherein the first opening is located at an end of a side surface of the coil assembly, the end of the side surface being adjacent to the head portion.
9. The ignition coil according to claim 8, wherein the second opening is located in the bottom surface of the head portion, and the air passage further includes a first groove located in a bottom surface of the coil assembly and connected to the first opening, a second groove located in a top

surface of the head portion and connected to the first groove, and a hole extending from a bottom of the second groove to the second opening.

10. An ignition coil-internal combustion engine assembly comprising: the ignition coil according to claim 1; and an internal combustion engine including a plug hole in which the tubular portion of the protector of the ignition coil is inserted.
